Syllabus Details

Syllabus ID:	14487
Syllabus Name:	**Interdisciplinary Business - Software Engineering Graduation Thesis_Khóa luận tốt nghiệp liên ngành Quản trị kinh doanh - Kỹ thuật phần mềm**
Course Name English:	**Interdisciplinary Business - Software Engineering Graduation Thesis**
Subject Code:	**GRC490**
Learning-Teaching Method:	Co-supervision; project-based learning; coaching
NoCredit:	10
Degree Level:	Bachelor
Time Allocation:	Study hours (500h) = 33.75 contact hours (135 minutes × 15 slots) + 1.5h defense + 464.75h self-study
Pre-Requisite:	A. Business track (BBA): Pass OJB202 (Vietnamese students); pass 80% of total credits (PHE_COM* and OJB202 excluded for Vietnamese students; PHE_COM* excluded for international students); complete (GPA>=4) at least one introductory technology/digital-tools course (e.g., EXE101, or equivalent); Complete concentration courses (GPA>=4): BBA_MKT: MKT201, MKT202, MKT301, MKT304; DMS301m/DMS302m, DMA301m/DMA302m BBA_IB: ECO201, IBI101, SCM201, IBS301m, RMB301, IBF301 BBA_HM: -HOM202, EVN201, HOM301c from K15-K20C; -HMO102, EVP201, HOM301c, RMB301, (LOD213 hoặc LOD222) và (RES223 hoặc FBO201) from K20C BBA_FIN: -FIN301, FIN303, FIN402, FIN403, ACC302 and (RMB301 or RMB302) for 2 combos Investment Finance & Corporate Finance -ACC305, FIN301, FIN303, FIN402, ACC302 and (RMB301 or RMB302) for other combos BBA_TM: TTG201, TTM201, TTM202, TTM203 B. Software Engineering track (SE): Pass PMG201c or PMG202c; SWD391 or SWD392; SWP391; pass 60% of the selected combo; pass OJT202 (not applicable to international students); pass 80% of total credits (PHE_COM* and OJT excluded); GPA ≥ 4 for each prerequisite course. C. Cross-disciplinary: team includes both tracks; recommended 6–7 students (3–4 SE + 2–3 BIZ), absolute minimum 2 SE + 2 BIZ.
Description:	GRC490 is an integrative capstone experience bridging Business Administration and Software Engineering. Interdisciplinary teams identify a genuine market problem, validate the business opportunity, design and build a functioning, deployable digital product, and defend both its business viability and technical implementation. Each team is guided by two co-supervisors—one from each faculty—and each student is assessed individually before a mixed committee.
StudentTasks:	1. Form an interdisciplinary team and select a project requiring both a business case and a working software product. 2. Divide and schedule work clearly across business and technical roles. 3. Attend at least 80% of scheduled supervisor meetings to remain eligible for defense. 4. Conduct primary and secondary market research. 5. Produce all required reports and a functioning, deployable software product through iterative development and testing. 6. Defend the business viability and technical implementation individually before the evaluation committee. 7. Academic dishonesty disqualifies the entire team from defense.
Tools:	Business analysis: market research platforms, Excel/Google Sheets, survey tools. Design: Figma, draw.io, Lucidchart or equivalent. Development: appropriate IDE, Git/GitHub, CI/CD pipeline and deployment platform. Project management: Jira, Trello, Notion or equivalent.
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	1028/QĐ-ĐHFPT dated 08/21/2026
IsApproved:	**True**
Note:	Delivered and assessed entirely in English. All business reports, technical documents, thesis, slides and oral defense must be in English. Technical documents should follow applicable international software-engineering standards (IEEE 830, IEEE 1016 or equivalent). Recommended team size: 6–7 students (3–4 SE + 2–3 BIZ).
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	8/21/2026

5 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Student's Guide for Interdisciplinary Graduation Thesis	—	FPT University				False	False	False	Developed by FPT University and issued with this syllabus
2	Quy chế đào tạo hệ đại học chính quy tại Trường Đại học FPT	—	FPT University	2022			False	False	False	Issued with Decision 774/QĐ-ĐHFPT dated 07/09/2022
3	Research Methods for Business Students	Saunders, M.; Lewis, P.; Thornhill, A.	Pearson Education Limited			978-0-273-75075-8	False	False	False	Recommended reference for business research methodology
4	Thực hành nghiên cứu trong kinh tế và quản trị kinh doanh	Nguyễn Văn Thắng	NXB Đại học Kinh tế Quốc dân				False	False	False	Recommended Vietnamese reference
5	Project Templates: SRS, SDD, STD, Financial Model and Business Plan	—	FPT University				False	False	False	Available on CMS/FLM; developed by FPT University

7 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Demonstrate an ability to integrate cross-disciplinary knowledge (business and technology), handle complexity, and plan and carry out advanced tasks independently and as a group member within specified time limits.
2	CLO2	Use appropriate market research methodology to identify market problems, validate demand, and propose a digital solution that is both technically feasible and commercially viable.
3	CLO3	Design, implement, and test a software system aligned with stated business requirements, using sound software engineering principles: requirements, architecture, iterative development, and testing.
4	CLO4	Develop critical thinking, creative problem-solving, and management capabilities; analyze marketing, operational, and financial aspects of the digital product in theory and practice.
5	CLO5	Apply project-management principles to coordinate an interdisciplinary team and deliver a working digital product within schedule and quality constraints.
6	CLO6	Write, present and defend the thesis—including the business case and technical design—according to well-defined academic and professional criteria, with correct referencing.
7	CLO7	Produce a feasible, deployable digital product or near-production MVP that demonstrates real-world entrepreneurial and engineering value and supports actual or potential business operation.

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Report 1: Project Introduction & Digital Opportunity Problem, digital gap, solution overview, team roles, feasibility	Offline	CLO1, CLO2, CLO7	IU	Syllabus; Student's Guide; Brainstorming template; Report 1 template	GRC490_Huong dan sinh vien	Finalize topic and roles; prepare Report 1	
2	Report 1 discussion & submission Project Management Plan kick-off; WBS; sprint/milestone schedule	Offline	CLO1, CLO2, CLO5	U	Syllabus; Report 2 Project Management Plan template		Submit Report 1; build integrated plan; begin Project Management Plan	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
3	Report 2A: Market Analysis & Value Proposition Market, competitors, segmentation, value proposition, revenue model	Offline	CLO1, CLO2, CLO7	IU	Student's Guide; Market Research Plan; Report 2A template		Execute primary and secondary research; begin Report 2A	
4	Report 2B: Software Requirements Specification (SRS) Functional/non-functional requirements; use cases; user stories	Offline	CLO1, CLO2, CLO3	IU	Student's Guide; SRS template		Write SRS using validated user needs; prepare Report 2B	
5	Reports 2A & 2B refinement Finalize market analysis, SRS, use cases and user journeys	Offline	CLO2, CLO3	U	Student's Guide; Reports 2A and 2B		Revise both reports; finalize approved use-case list	
6	Submit Reports 2A & 2B Report 3A: Software Architecture & Design (SDD), database, API and UX/UI	Offline	CLO3, CLO5	IU	SDD template; UI/UX design guide		Submit 2A/2B; begin design and UI prototype; prepare 3A	
7	Report 3B: Key Activity & Go-to-Market Plan Operations, organization, marketing, KPIs and revenue execution	Offline	CLO4, CLO5, CLO7	IU	Marketing Plan template; Student's Guide		Draft 3B; map operational KPIs to product metrics	
8	Submit Reports 3A & 3B Development Iteration 1: core features, coding, unit testing and demo	Offline	CLO3, CLO5	U	Test Documentation template; Weekly Progress Report		Submit 3A/3B; complete Iteration 1 and demo	
9	Development Iteration 2 Expanded features, integration testing, customer-journey validation and demo	Offline	CLO3, CLO4, CLO5	U	Test Documentation; Weekly Progress Report		Complete Iteration 2; update SRS/SDD; validate against market requirements	
10	Development Iteration 3 Feature completion, regression testing, STD preparation and demo	Offline	CLO3, CLO5	U	STD template; Weekly Progress Report		Complete Iteration 3; draft STD; update technical documents	
11	System Testing, Quality Assurance & Business Validation Full test execution, defect fixing and STD submission	Offline	CLO3, CLO4, CLO5	U	STD; QA checklist		Execute system tests; fix defects; submit STD; validate product–market fit	
12	Report 4: Financial Analysis & Conclusions Startup cost, revenue, P&L, cash flow, break-even and 3-year projections	Offline	CLO4, CLO7	IU	Financial Model template; Student's Guide		Draft Report 4; integrate hosting, DevOps, license and maintenance costs	
13	Report 4 refinement Business/technical risk register, backup plans, scaling roadmap and conclusions	Offline	CLO4, CLO7	U	Risk Register template; Student's Guide		Complete risks, roadmap and Report 4	
14	Submit Report 4, User Guides, Final Software Package and Thesis Defense preparation and Q&A practice	Offline	CLO6, CLO7	U	User Guide template; Thesis compilation guide		Submit complete thesis and deployed product; prepare presentation and Q&A	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
15	Defense 30-minute presentation; 20-minute product demo; 40-minute Q&A and committee meeting	Offline	CLO6	U	Final thesis; deployed software; presentation slides		Present and answer individually	

0 Constructive question(s)

10 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Final Thesis Document + Software Package	on-going	1	10.0%	5	Submit by Session 14	CLO6, CLO7	Compiled thesis, guides and deployed product		Coherence, writing, references, product completeness and deployment readiness	Evaluated by co-supervisors and reviewed by the committee	Product mus be functiona deployed ar free of show stopper defe
2	Project Management	on-going	1	5.0%	4	Assessed continuously in Sessions 2–14	CLO1, CLO5	Plan, progress records and tool evidence		WBS, sprint tracking, role balance, version control, attendance, scope and risk handling	Graded jointly by both co-supervisors; individual mark = 0%–120% of group mark; team average ≤ group mark	Mark differe >2/10 requir reconciliatio
3	Report 1: Project Introduction & Digital Opportunity	on-going	1	5.0%	4	Submit by Session 2	CLO1, CLO2, CLO7	Written project report		Problem, digital gap, product concept and team-role division	Graded by co-supervisors; individual mark = 0%–120% of group mark based on contribution; team average ≤ group mark	Required for course progression
4	Report 2A: Market Analysis & Value Proposition	on-going	1	5.0%	4	Submit by Session 6	CLO1, CLO2, CLO7	Written market-analysis report		Research quality, segmentation, competition, revenue model and value proposition	Graded by co-supervisors; individual mark = 0%–120% of group mark	Evidence fro primary and secondary research required
5	Report 2B: Software Requirements Specification (SRS)	on-going	1	5.0%	4	Submit by Session 6	CLO1, CLO2, CLO3	SRS document		Use cases, transactions, non-functional requirements and alignment with business needs	Graded by co-supervisors using the SRS rubric	Minimum SR grade 2/10 required for defense eligibility
6	Report 3A: Software Design Document (SDD) & UI/UX	on-going	1	5.0%	4	Submit by Session 8	CLO3, CLO5	SDD and UX/UI prototype		Architecture, database, API and UI/UX alignment with user needs	Graded by co-supervisors using the SDD rubric	Minimum SD grade 2/10 required

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
7	Report 3B: Key Activity & Go-to-Market Plan	on-going	1	5.0%	4	Submit by Session 8	CLO4, CLO5, CLO7	Written business plan component		Operations, HR, marketing plan and KPI linkage to product capabilities	Graded by co-supervisors; individual mark = 0%–120% of group mark	
8	Report 4: Financial Analysis, Risk & Roadmap	on-going	1	5.0%	4	Submit by Session 14	CLO4, CLO7	Financial model and written report		Tech-inclusive financial model, risk register and feasible product roadmap	Graded by co-supervisors; individual mark = 0%–120% of group mark	
9	Software Testing Documentation (STD)	on-going	1	5.0%	4	Submit by Session 11	CLO3, CLO5	STD and test evidence		Test plan, coverage, defect tracking and system-test results	Graded by co-supervisors using the STD rubric	Minimum ST grade 2/10 required
10	Oral Defense: Business Plan Presentation & Product Demo	Final exam	1	50.0%	4	90 minutes: 30-minute presentation + 20-minute demo + 40-minute Q&A/committee	CLO6, CLO7	Presentation, deployed-product demo and individual Q&A		Business feasibility, technical quality, interdisciplinary integration, presentation and individual mastery	Each examiner scores each student individually. Group criteria: 35/50; individual Q&A: 15/50. Convert rubric total to 10-point scale by dividing by 5.	Q&A graded examiners o the student' faculty; interdisciplir criteria grad jointly